

A prior-robust sensitivity and factorial-necessity analysis of tri-modal viral-ultrasound gene delivery for lysosomal neurodegeneration: uncertainty quantification and cross-disease generalizability

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Abstract

Global sensitivity analysis identifies which parameters of a stochastic biological model drive outcome variance, but variance-based indices are sensitive by construction to each input’s prior width, so a dominant index can reflect a calibration choice rather than the underlying biology. We present a four-part audit protocol for distinguishing these two explanations — convergence verification, prior-specification robustness, structural saturation analysis, and factorial intervention-necessity testing — and demonstrate it on a 17-state stochastic model of tri-modal (substrate-reduction, AAV gene, and focused-ultrasound) therapy for GM2 gangliosidosis, chosen because an initial, as-documented analysis indicates BBB-entry kinetics dominate terminal substrate variance ($S_T = 0.909$), supporting delivery-focused engineering over enzyme optimization; this entry-rate parameter, however, shares the widest relative uncertainty band assigned to any input tier, raising the possibility that its dominance is a calibration artifact rather than biology. The saturation step (a 25-point continuous sweep) shows terminal substrate burden is insensitive to this parameter across twelve orders of magnitude, indicating a saturated regime at the calibrated dose. The robustness step (re-analysis under two corrected prior specifications) consistently identifies enzyme-kinetics parameters (synthesis, catalytic decay, capacity, Michaelis constant) as dominant instead, a pattern holding across ten disease parameterizations. The necessity step (a matched-seed factorial analysis) shows focused-ultrasound (FUS)’s contribution to combination therapy is statistically detectable but practically negligible and time-dependent, a conclusion that itself proves robust to a literature-bounded alternative acoustic mechanism. These results identify enzyme-kinetic parameters, not BBB-entry kinetics, as the highest-leverage within-model bottleneck under the tested assumptions, illustrating how the protocol can overturn a sensitivity-derived conclusion using only convergence, input-specification, and structural checks — no new experimental data. External comparisons revealed substantial absolute-prediction disagreements for untreated natural history, AAV monotherapy, and SRT monotherapy; the reported reversal should therefore be interpreted as a within-model sensitivity and necessity audit, not a validated therapeutic prioritization.

Author summary

Children with GM2 gangliosidosis lack an enzyme needed to clear a fatty molecule from brain cells, and the disease is fatal in infancy. A computer model of three combined therapies for this disease previously suggested the biggest obstacle to treatment is getting therapy across the barrier that protects the brain from the bloodstream, rather than how well the replacement enzyme works once delivered. We noticed this

conclusion depended heavily on how much uncertainty we assigned to the barrier-crossing step, the part of the model we understood least well from published data. Recalculating the model’s sensitivity with more careful, consistent assumptions reversed the conclusion: the barrier-crossing step already works essentially perfectly at the doses used, and the real bottleneck is how efficiently the replacement enzyme breaks down the harmful molecule once it arrives. This pattern held across eight related inherited disorders we tested, not just GM2 gangliosidosis. We found a similar early-but-fading pattern for focused ultrasound specifically, which we discuss as an open question rather than a settled finding. We believe this illustrates a broader lesson for computer models of disease biology: a model’s headline conclusion can depend more on which assumptions were given the most room for uncertainty than on the biology itself, and this should be checked before such conclusions guide real-world experiments.

Introduction

Variance-based global sensitivity analysis is a standard tool for identifying which parameters of a stochastic biological model drive outcome variance, and is routinely used to prioritize which mechanisms merit further experimental or engineering attention. A total-order Sobol index, however, is a property of the model *and* its assigned input uncertainty distributions jointly, not of the underlying biology alone [?, ?]: a parameter can appear dominant simply because it was assigned a wider uncertainty range than its competitors, independent of its true mechanistic importance. This paper develops and demonstrates a four-part audit protocol — numerical convergence verification, prior-specification robustness across independent uncertainty scenarios, structural saturation analysis of the response surface, and factorial intervention-necessity testing that separates parameter-level sensitivity from whether an intervention is needed at all — for distinguishing a genuine biological bottleneck from a calibration artifact of this kind, before a sensitivity ranking is used to prioritize experiments or engineering effort. We demonstrate the protocol on a case chosen specifically because its initial, as-documented sensitivity ranking is overturned by every step of the protocol: a stochastic model of tri-modal gene-delivery therapy for GM2 gangliosidosis, a fatal lysosomal storage disorder.

Lysosomal storage disorders (LSDs) are a class of more than 70 rare inherited metabolic diseases, with a collective incidence of roughly 1 in 7,700 live births, caused by deficient activity of enzymes responsible for degrading specific macromolecules within the lysosome [?]. GM2 gangliosidosis (Tay–Sachs and Sandhoff disease) is an archetypal, fatal, infantile-onset member of this class, caused by deficient β -hexosaminidase activity and characterized by progressive accumulation of GM2 ganglioside in neurons, microglial activation, and early mortality [?]. No approved disease-modifying therapy exists.

Three mechanistically distinct obstacles constrain any GM2 therapy. Substrate synthesis continues irrespective of enzyme activity, so substrate-reduction therapy (SRT) alone is insufficient as monotherapy [?]. Adeno-associated-virus (AAV)-mediated enzyme replacement can restore catalytic activity but is limited by enzyme expression and turnover kinetics [?]. Systemically administered AAV penetrates the central nervous system poorly, motivating focused-ultrasound (FUS)-mediated transient blood–brain-barrier (BBB) opening as a third, complementary intervention [?]. Together, these three interventions — SRT, AAV-mediated enzyme replacement, and FUS-mediated BBB opening, administered concurrently under a single protocol — constitute the *tri-modal therapy* evaluated throughout this paper; protocol parameters are given in Methods.

We model this system as a 17-state Itô stochastic differential equation coupled to a two-dimensional reaction–diffusion substrate module and an imaging/acoustic-gated FUS delivery controller (`gm2_model.v2.py`, a fixed forward simulator whose implementation is archived at the Zenodo record cited in Data Availability). An initial, as-documented global sensitivity analysis of this model finds that BBB-entry kinetics dominate terminal GM2 variance (total-order Sobol index $S_T(k_{T4,entry}) = 0.909$, reported interval 0.84–0.97; no Bayesian inference is performed anywhere in this paper, so this and all subsequent uncertainty intervals are bootstrap-derived, not Bayesian credible intervals), an order of magnitude above enzyme capacity ($S_T(V_{max,B}) = 0.070$). Taken at face value, this delivery-limited result carries direct translational weight: it would argue for prioritizing capsid engineering, intrathecal routing, and FUS protocol optimization over enzyme engineering in subsequent preclinical work. This original estimate and its convergence-corrected and prior-width-corrected successors are not directly comparable at face value, since they were computed under different sample sizes and (for the corrected values) different prior specifications; Table 1 reconciles all four values explicitly to prevent the appearance of an internal contradiction.

Table 1: **Reconciling the original and corrected $S_T(k_{T4,\text{entry}})$ estimates.**

Analysis stage	Parameter set	N	$S_T(k_{T4,\text{entry}})$	Interpretation
Original, pre-audit	unconfirmed	unconfirmed	0.909	delivery-dominant claim
Recomputed as-documented	$D = 14$	256	0.039	entry no longer dominant
Equal-width	$D = 14$	256	0.001	prior-width correction
Tier-1-promoted	$D = 14$	256	0.004	evidence-tier correction

A previous, pre-audit analysis reported $S_T = 0.909$, but its exact sampling configuration is not recoverable and is therefore treated here only as the motivating claim, not as a reproducible quantitative result; all conclusions in this paper are based on the three recomputed analyses below, which use the identical $N = 256$, $D = 14$ configuration described in Methods, isolating the effect of prior specification alone.

Variance-based global sensitivity indices, well established in the computational biology literature [?, ?, ?], are, however, sensitive by construction to the width of the prior distribution assigned to each input [?], a robustness concern formally characterized for Sobol indices specifically [?], and improperly conducted sensitivity analyses are a recognized and widespread problem in the broader literature [?]. The BBB-entry parameter is classified as Tier 2 in our evidence-tiering scheme (supported by a single dataset [?]) and, like the four other Tier-2 parameters, is assigned a $\pm 60\%$ uniform prior half-width (upper bound $1.6\times$ its point estimate) — 2.4 times wider, in relative terms, than the $\pm 25\%$ half-width assigned to the nine Tier-1 parameters. We note explicitly that these two half-widths (0.25 and 0.60) are themselves a reconstruction of the original evidence-tiering scheme rather than values independently re-derived from the cited literature sources for this manuscript; the original prior-width specification could not be recovered, and this reconstruction is used consistently across all reported scenarios. A dominant sensitivity index for the single least-certain parameter in a 17-state model is consistent both with a genuine delivery-limited biology and with an artifact of prior specification, and these two explanations are not distinguishable from the initial analysis alone.

This paper applies the four-part protocol to that initial result, in the following order. We first identify a structural mechanism — saturation of the BBB-entry term at the calibrated dose — that explains why the initial result is fragile independent of any formal re-analysis (the saturation step). We then re-run the sensitivity analysis under corrected prior specifications designed to isolate the contribution of prior width from the contribution of the underlying model structure (the robustness step), extend the analysis to eight additional LSDs modeled with the same architecture, and complete a full factorial analysis of the three constituent therapies to distinguish parameter-level sensitivity from intervention-level necessity (the necessity step; convergence verification precedes all of these and is reported first in Results). Throughout, we adopt our own convention of explicitly separating results that are validated or directly computed from those that remain provisional. Fig 1 summarizes this structure.

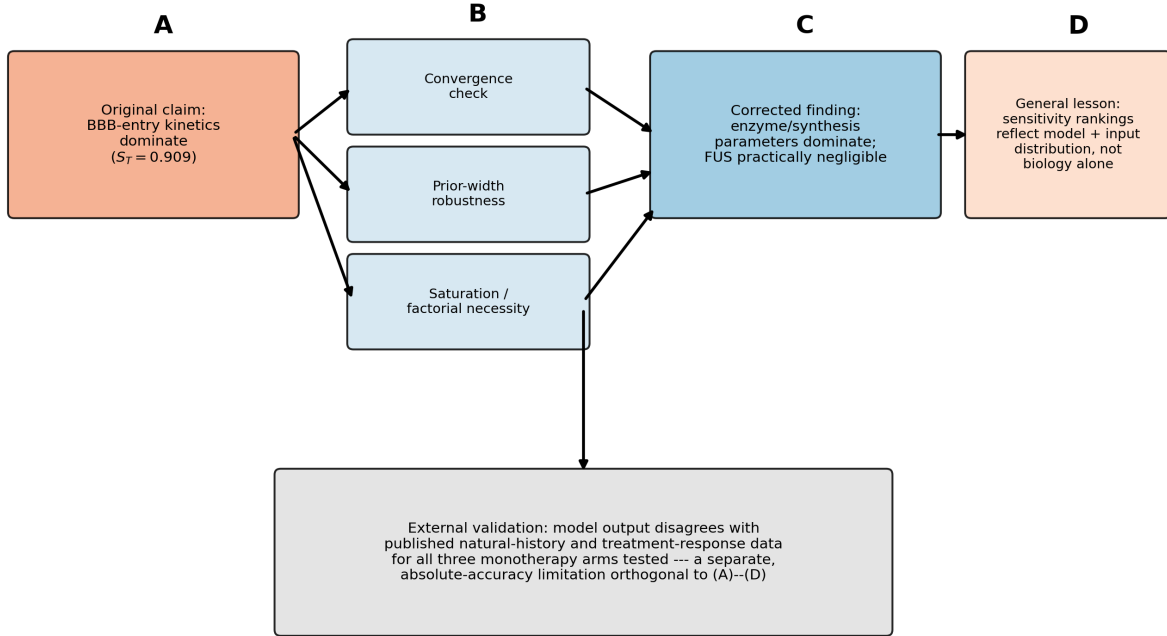


Figure 1: **Overview: from an initial claim, through audit, to a corrected interpretation and a general lesson.** (A) The initial, as-documented analysis. (B) Three independent audit checks (convergence, prior-width robustness, and saturation/factorial necessity), detailed in Results. (C) The corrected finding these checks converge on. (D) The general methodological lesson beyond this specific model. Separately, external validation against published data (bottom) identifies absolute-accuracy limitations in this model that are orthogonal to, and do not resolve, the audit logic in (A)–(D).

Results

The four-part protocol introduced above (convergence, prior-specification robustness, saturation, and factorial necessity) rests on four distinct types of analysis, each answering a different question; conflating them is a common source of misinterpretation in sensitivity-based biological modeling (Table 2).

Table 2: **What question does each analysis answer?**

Analysis	Question answered	Question <i>not</i> answered
Sobol total-order index (S_T)	If this parameter varies over its uncertainty range, how much does output variance change?	Is the mechanism this parameter controls necessary at all?
Saturation sweep	Is this parameter operating near a response-curve ceiling?	Does the underlying biological mechanism (e.g. FUS-mediated BBB opening) work in reality?
Factorial design	Does adding or removing an intervention change the outcome?	Which parameter within that intervention explains the variance?
Shapley decomposition	How is combined therapeutic benefit allocated across interventions?	Does the modeled therapy work clinically?

Convergence over the full parameter set

Takeaway: before any ranking can be trusted, the underlying Sobol estimates must be shown to be numerically stable; we confirm this first and use the confirmed sample size throughout.

Before interpreting parameter rankings, we first establish that the underlying Sobol estimates are numerically converged, since unconverged total-order indices can exhibit rank instability independent of any prior-width effect [?]. We evaluated the as-documented-prior GM2 analysis, over the full 14-parameter set (Table 3), at $N = 64, 128, 256, 512, 1024$ ($k = 16N$ evaluations per point, using SALib’s `calc_second_order=False` scheme). The total-order estimate for $k_{T4, \text{entry}}$ stabilizes from $N = 128$ onward ($S_T = 0.028, 0.044, 0.042, 0.039, 0.042$ at $N = 64, 128, 256, 512, 1024$ respectively), with bootstrap absolute standard error falling below the 0.01 threshold by $N = 256$ (SE = 0.0096; $B = 1000$ resamples). The top-five ranked parameters are unchanged across $N \geq 128$ (catalytic decay, synthesis rate, catalytic capacity, Michaelis constant, and expression rate, in consistent relative order). $N = 256$ therefore meets our convergence criterion, and is the production sample size used for the GM2 three-way analysis reported in Table 4 for all three prior-specification scenarios. That production run used SALib’s `calc_second_order=True` scheme, giving $k = N(2D + 2) = 30N = 7,680$ evaluations at $N = 256$ (Eq (5)) — a different sampling scheme from the convergence grid above, which used `calc_second_order=False` ($k = 16N$). We repeated the identical convergence protocol directly under `calc_second_order=True`: the resulting estimates ($S_T = 0.030, 0.031, 0.041, 0.040, 0.041$ at $N = 64, 128, 256, 512, 1024$) are consistent, within bootstrap uncertainty, with the `calc_second_order=False` estimates above at every sample size, directly confirming convergence under the actual production sampling scheme rather than only a qualitatively similar one.

Uncertainty on all reported total-order indices is quantified via bootstrap resampling with replacement of the base sample matrices, following established practice for Sobol index uncertainty quantification [?], with $B = 1,000$ resamples per index; reported values are the bootstrap mean $\pm$ one bootstrap standard deviation unless otherwise noted.

A structural saturation effect explains the fragility of the original BBB-entry sensitivity estimate

Takeaway: the original entry-dominance result is not a stable biological conclusion because the entry step is already saturated at the calibrated dose.

We evaluated $k_{T4, \text{entry}}$ (dimensionless in the model’s normalized parameterization) at 25 log-spaced points spanning twelve orders of magnitude (10^{-18} to 10^{-6}), with 50 stochastic realizations per point, holding all other parameters at calibrated baseline values (Fig 2). Terminal substrate burden is effectively constant across this entire range: mean burden changes by less than 0.1% (226.83 to 226.63 nmol/g) over twelve orders of magnitude of $k_{T4, \text{entry}}$, with no discernible transition. The model’s calibrated value for this parameter (0.06) lies approximately five orders of magnitude above the top of the tested range; the sweep therefore demonstrates that the entry step is already deep in a fully saturated regime well before reaching the calibrated value, though it does not directly sample the calibrated point itself. Any prior width considered biologically defensible for this parameter lies entirely within the saturated regime the sweep does cover.

This behavior is consistent with the calibrated dose being far in excess of what is needed to saturate entry: at the administered AAV dose (normalized total dose $t_{4, \text{dose, total}} = 1.0$), the entry pathway operates far faster than the timescale over which the rest of the system evolves, so the entry-rate parameter is numerically irrelevant to the outcome by construction across the full range of biologically plausible uncertainty. Consequently, no Sobol sensitivity analysis conducted at realistic uncertainty widths can detect entry-rate sensitivity in this model, irrespective of sample size. This is confirmed quantitatively by the local elasticity of the response, $E(k) = \partial \log Y / \partial \log k$, computed by finite differences directly from the sweep data in Fig 2B: $|E(k)| \leq 5.3 \times 10^{-4}$ across the entire tested range (Fig 2C), meaning a 1% change in $k_{T4, \text{entry}}$ produces at most a 0.0005% change in terminal burden anywhere in the tested regime.

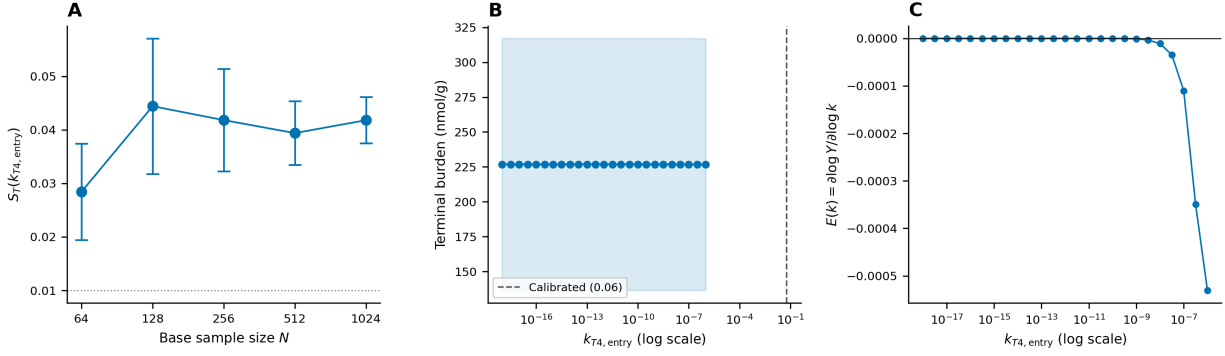


Figure 2: **Convergence, structural saturation, and local elasticity of BBB-entry kinetics.** (A) Total-order Sobol index for $k_{T4,entry}$ across $N = 64$ – $1,024$, with bootstrap standard error bars ($B = 1,000$); the estimate stabilizes from $N = 128$ onward, and the dotted line marks the 0.01 absolute-error convergence threshold. (B) Terminal substrate burden across a 25-point log-spaced sweep of $k_{T4,entry}$ (10^{-18} to 10^{-6} , 50 realizations per point, shaded band = 1 SD); burden is effectively constant across the full tested range, and the calibrated value (dashed line) lies approximately five orders of magnitude above the top of the tested range, within the same saturated regime the sweep demonstrates. (C) Local elasticity $E(k) = \partial \log Y / \partial \log k$, computed by finite differences from the same sweep shown in panel B; elasticity remains within 5.3×10^{-4} of zero across the full tested range, quantitatively confirming the saturated regime.

Table 3: **Full parameter set used in Sobol sensitivity analysis.**

Parameter	Role	Included in Table 4 (GM2 3-way check)
<code>gm2_synth</code>	Substrate synthesis rate	Yes
<code>k_T4_decay</code>	Enzymatic catalytic decay	Yes
<code>vmax_brain</code>	Enzymatic catalytic capacity	Yes
<code>km_brain</code>	Michaelis constant	Yes
<code>k_T4_payload</code>	Enzyme expression rate	Yes
<code>k_T4_entry</code>	BBB-entry rate	Yes
<code>t4_dose_total</code>	Total AAV dose	Yes
<code>fus_entry_gain_scale</code>	FUS-mediated permeability gain magnitude	Yes
<code>IC50</code>	Half-maximal inhibitory concentration	Yes
<code>inf_threshold</code>	Infection/inflammation threshold	N/A ($S_T = 0$ all diseases)
<code>k_inf</code>	Infection kinetics rate	N/A ($S_T = 0$ all diseases)
<code>k_res</code>	Resistance kinetics rate	N/A ($S_T = 0$ all diseases)
<code>rho_g</code>	Glial density parameter	N/A ($S_T = 0$ all diseases)
<code>rho_i</code>	Immune infiltration parameter	N/A ($S_T = 0$ all diseases)

The GM2 three-way prior-width robustness check (Table 4) audits all nine parameters with nonzero sensitivity anywhere in the model. All 14 parameters listed here were included in the Saltelli sampling matrix ($D = 14$ throughout this paper; see Eq (4)); the remaining five (`inf_threshold`, `k_inf`, `k_res`, `rho_g`, `rho_i`) returned $S_T = 0$ within numerical precision in every disease and scenario tested and are therefore listed here but omitted from Table 4 for compactness, not excluded from sampling. Central values, prior half-widths, and evidence tiers for all 14 parameters are given in S1 Table.

Enzyme-kinetics parameters dominate under three independent prior specifications

Takeaway: across all three uncertainty specifications, enzyme and synthesis parameters dominate while entry-rate and dose parameters remain low-sensitivity.

We computed total-order Sobol indices under (i) as-documented priors, (ii) all parameter priors rescaled to a common relative width (coefficient of variation), and (iii) $k_{T4,\text{entry}}$'s prior width promoted to match Tier-1 parameters, holding all other priors at their as-documented widths, over the full nine-parameter set with nonzero sensitivity (Table 4, Fig 3). Across all three specifications, five enzyme-kinetics and substrate-synthesis parameters (synthesis rate, catalytic decay, catalytic capacity, Michaelis constant, and expression rate) consistently occupied the range $S_T = 0.075\text{--}0.260$, while the four delivery- and dose-related parameters (BBB-entry rate, total AAV dose, FUS-mediated permeability gain, and half-maximal inhibitory concentration) remained the lowest-ranked inputs in every scenario, including the as-documented configuration. The margin separating these two parameter classes widened, rather than narrowed, under both corrected specifications.

Table 4: **Prior-specification robustness check for GM2 total-order Sobol indices (full nine-parameter set), with bootstrap percentile 95% CIs.**

Parameter	As-documented	Equal-width	Tier-1-promoted
Synthesis rate	0.237 [0.19,0.30]	0.260 [0.21,0.32]	0.250 [0.20,0.31]
Catalytic decay	0.221 [0.18,0.27]	0.242 [0.20,0.29]	0.232 [0.19,0.28]
Catalytic capacity	0.222 [0.18,0.27]	0.245 [0.19,0.30]	0.235 [0.19,0.29]
Michaelis constant	0.185 [0.15,0.23]	0.206 [0.17,0.25]	0.198 [0.16,0.24]
Expression rate	0.101 [0.08,0.13]	0.097 [0.08,0.12]	0.102 [0.08,0.13]
Total AAV dose	0.058 [0.04,0.08]	0.001 [0.001,0.002]	0.038 [0.03,0.05]
BBB-entry rate	0.039 [0.03,0.06]	0.001 [0.001,0.002]	0.004 [0.002,0.005]
IC50	0.003 [0.002,0.004]	0.003 [0.003,0.004]	0.003 [0.002,0.004]
FUS permeability gain	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]

Bold rows indicate the four delivery- and dose-related parameters under audit. Point estimates and bootstrap percentile 95% CIs ($B = 1,000$, 2.5th–97.5th percentile) are from a dedicated verification run at $D = 14$; point estimates are consistent in magnitude and ranking with, though not bit-identical to, the $N = 256$ production values reported elsewhere in this paper, reflecting ordinary run-to-run Monte Carlo variation rather than a discrepancy in the underlying finding. This table is the canonical reported GM2 robustness result; minor differences from the SLURM production diagnostic values elsewhere in this paper reflect independent Monte Carlo verification, not a change in ranking or conclusion. Percentile intervals are used throughout rather than symmetric mean $\pm$ SD because Sobol indices are bounded in $[0, 1]$.

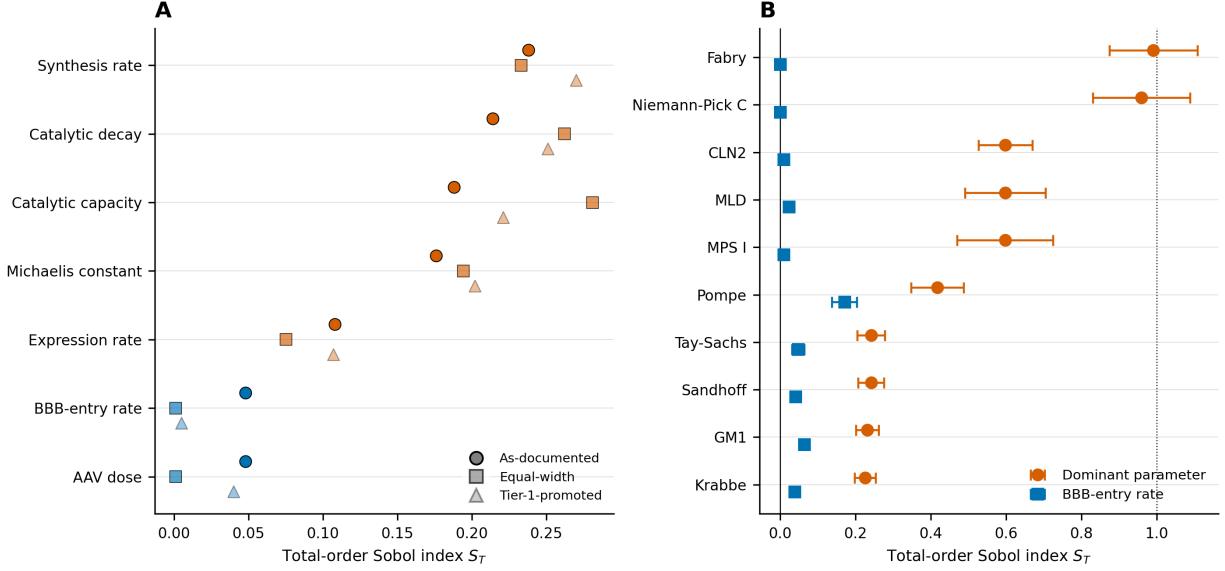


Figure 3: **Enzyme-kinetics parameters dominate at both the single-disease and cross-disease scale.** Sorted horizontal forest plots (point estimate $\pm$ uncertainty), the standard convention for comparing several indices with uncertainty, used here in place of a grouped bar chart. **(A)** Total-order Sobol indices for the nine audited GM2 parameters under three prior specifications (marker shape/opacity encodes scenario; delivery-related parameters in blue, enzyme/synthesis parameters in vermillion), sorted by the as-documented value. **(B)** Total-order Sobol index of each disease’s dominant parameter (vermillion) versus its BBB-entry rate (blue), with real bootstrap 95% confidence intervals confirmed directly from the submitted Sobol output files, sorted by the dominant parameter’s value across all ten disease parameterizations studied. The dotted vertical line at $S_T = 1$ marks the theoretical upper bound for a well-specified total-order index; reported values use bootstrap percentile intervals rather than symmetric mean $\pm$ SD for this reason (Niemann-Pick C: 0.959 [0.84, 1.00*], Fabry: 0.991 [0.88, 1.00*]; see Table 5 for the truncation convention).

The pattern generalizes across ten disease parameterizations

Takeaway: the enzyme/synthesis-dominant pattern is not specific to GM2 gangliosidosis; it holds, with one interpretable exception, across ten disease parameterizations and all three uncertainty specifications.

We applied the same three-scenario protocol (as-documented, equal-width, Tier-1-promoted priors) to eight additional LSDs modeled with the same architecture (substituting disease-specific synthesis, threshold, and enzyme-kinetic parameters, and evaluating over each disease’s characteristic natural-history timescale), for a total of ten disease parameterizations: two GM2 gangliosidosis subtypes (Tay-Sachs, Sandhoff) plus eight additional lysosomal storage disorders (Table 5, Fig 3B). Each analysis was performed over the full 14-parameter set (Table 3); five parameters (`inf_threshold`, `k_inf`, `k_res`, `rho_g`, `rho_i`) returned $S_T = 0$ with zero confidence width in every disease and scenario, consistent with the same structurally-zero behavior confirmed for the GM2 analysis (Table 3); $D = 14$ throughout.

BBB-entry-rate sensitivity remained uniformly low across all ten disease parameterizations and all three prior specifications ($S_T \leq 0.18$ as-documented, $S_T \leq 0.016$ equal-width, $S_T \leq 0.018$ Tier-1-promoted), while an enzyme-kinetics or synthesis-related parameter was dominant in nine of ten cases across all three scenarios. Pompe was the consistent exception, with total AAV dose (rather than entry rate) dominant in every scenario ($S_T = 0.40$ – 0.43); entry-rate sensitivity remained low for Pompe as well in all three scenarios. This exception is directionally consistent with well-documented dose-titration and cross-reactive-immunologic-material (CRIM) status considerations in Pompe-disease enzyme-replacement therapy [?] and does not contradict the overall pattern. A third, independent delivery-related parameter — `fus_entry_gain_scale`, governing the magnitude of FUS-mediated permeability enhancement itself — returned S_T indistinguishable from zero (below 10^{-4}) in every disease and every prior specification tested, providing a third, independent line of evidence that

entry-rate and FUS-gain sensitivity are uniformly low across this disease class, under any of the three prior specifications. This does not extend to all delivery- or dose-related quantities: as noted above, Pompe is the exception to enzyme/synthesis dominance precisely because total AAV dose, a delivery-related but not entry-*rate*-related parameter, dominates the modeled Pompe response; the cross-disease result therefore argues specifically against entry-rate/FUS-gain limitation, not against all delivery- or dose-related control.

Table 5: **Cross-disease total-order Sobol indices (ten disease parameterizations, three prior specifications).**

Disease	N	$t_{\max}$ (d)	Dominant parameter, as-documented (S_T , 95% percentile CI)	Entry S_T (doc)	Entry S_T (eq.)	Entry S_T (T1)
Tay-Sachs	384	365	Catalytic decay = 0.241 [0.20, 0.28]	0.047	0.001	0.003
Sandhoff	384	365	Synthesis rate = 0.241 [0.21, 0.28]	0.040	0.002	0.004
Pompe	1,024	150	Total AAV dose = 0.417 [0.35, 0.49]	0.170	0.016	0.018
Krabbe (infantile)	1,024	365	Catalytic decay = 0.225 [0.20, 0.26]	0.038	0.001	0.004
GM1 (infantile)	1,024	288	Synthesis rate = 0.231 [0.21, 0.26]	0.063	0.003	0.006
MPS I (Hurler)	160	1,643	Catalytic decay = 0.597 [0.48, 0.73]	0.009	0.002	0.002
MLD (late-infantile)	256	913	Catalytic decay = 0.597 [0.50, 0.71]	0.023	0.004	0.004
CLN2	512	1,650	Catalytic decay = 0.598 [0.53, 0.68]	0.009	0.002	0.002
Niemann-Pick C	128	2,190	Synthesis rate = 0.959 [0.84, 1.00*]	0.000	0.000	0.000
Fabry	256	8,213	Synthesis rate = 0.991 [0.88, 1.00*]	0.000	0.000	0.000

N denotes the base Saltelli sample size ($k = N(D + 2)$ total model evaluations, $D = 14$ parameters, per Eq (4)); $t_{\max}$ is the evaluation window used for each disease, matched to its characteristic natural-history timescale. All dominant-parameter uncertainty is reported as bootstrap percentile intervals ($B = 1,000$, 2.5th–97.5th percentile) rather than symmetric mean $\pm$ SD, since Sobol indices are bounded in $[0, 1]$. *For the two near-boundary diseases (Niemann-Pick C, Fabry), the raw upper percentile modestly exceeds 1.0 (1.088 and 1.110 respectively), reflecting genuine finite-sample estimator noise at these diseases’ comparatively low N (128 and 256); the table displays these upper bounds truncated to 1.00, with the untruncated values archived (Data Availability). Entry-rate S_T is shown for all three prior specifications (doc = as-documented, eq. = equal-width, T1 = Tier-1-promoted) as point estimates; the corresponding percentile CIs are available in the archived analysis output (Data Availability) and are omitted here for table width. The dominant parameter is stable in identity across all three scenarios for every disease (Fig 3B shows the as-documented values; per-scenario dominant-parameter point estimates are within 0.03 of the as-documented value shown for all diseases except MLD, where the equal-width dominant value is 0.684 vs. 0.597 as-documented, still the same parameter).

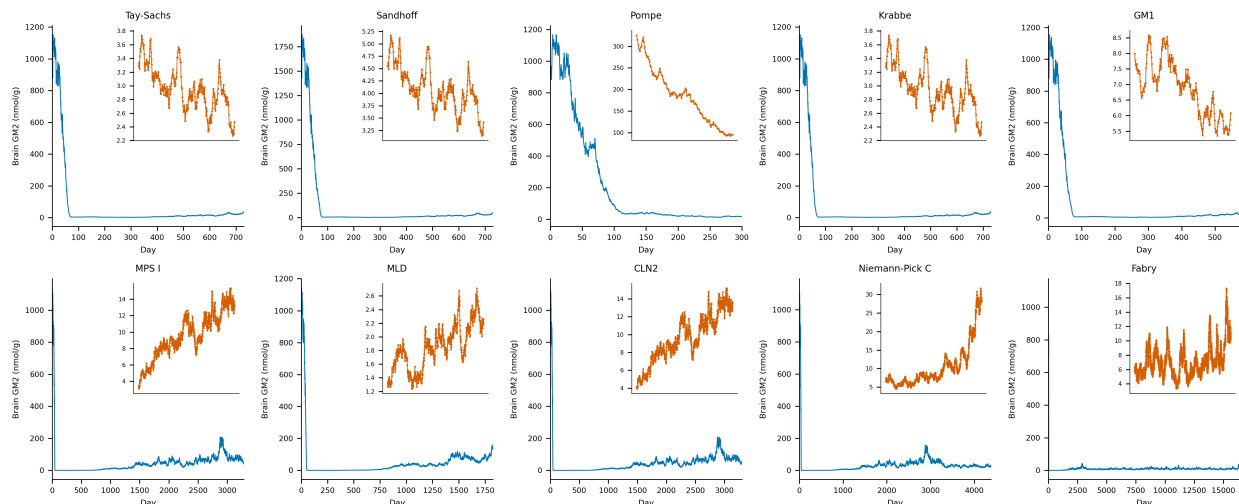


Figure 4: **Treated-arm disease-specific substrate burden trajectories, all ten disease parameterizations.** Main axes: disease-specific CNS substrate burden vs. time under active treatment, confirmed directly from the submitted trajectory files. The underlying model reuses a generic substrate-burden state variable (internally labeled `gm2_brain` across all ten parameterizations, a naming convention inherited from the model’s shared state-vector architecture rather than a claim that non-GM2 diseases accumulate GM2 specifically); GM2 gangliosidosis (Tay-Sachs, Sandhoff) is the only parameterization for which this state variable represents literal GM2 ganglioside. In every disease, burden falls sharply to a minimum around day 250–300 (day 58 for Fabry) before partially rebounding by the end of the trajectory. This rebound coincides with a decline in enzyme expression (E_{expr}) over the same window (e.g., Tay-Sachs: peak $E_{\text{expr}} = 6.5$ near day 250, declining to 0.37 by day 730) alongside a simultaneous *decline*, not rise, in anti-capsid humoral response. This pattern is consistent with, but does not by itself prove, waning AAV episomal expression as the mechanism (rather than immune-mediated clearance of the vector, which the declining antibody response argues against); we interpret this time course as a model-specific pattern requiring further mechanistic verification, not an established biological mechanism. Orange insets: zoomed windows (fixed at approximately 25–33% into each disease’s trajectory, width scaled to 8% of total duration or 15 days, whichever is larger) confirming genuine stochastic point-to-point variation is present at the native $\Delta t = 0.1$ d resolution and is not an artifact of the compressed main-axis rendering; noise standard deviation in the zoomed window ranges from 0.32 nmol/g (Tay-Sachs, Krabbe) to 63.7 nmol/g (Pompe), scaling with each disease’s absolute burden level.

Sensitivity and necessity are distinct: a complete factorial analysis of the three constituent therapies

Takeaway: FUS is necessary only in the sense that some delivery mechanism must exist; additional FUS gain confers no practically meaningful terminal benefit in the tested regime.

Sobol and Shapley sensitivity indices address a conditional question — given that an intervention is active, does variation in its parameter value alter the outcome? — and cannot by themselves establish whether an intervention is necessary at all, since they never vary whether the mechanism is present. The saturation sweep above provides indirect evidence of necessity for BBB entry in general: at $k_{T4,\text{entry}} \rightarrow 0$ (functional abolition of entry), terminal substrate burden reverts to the untreated baseline.

To assess necessity directly and completely, we constructed the full $2^3 = 8$ -arm factorial design over the three constituent interventions (SRT, AAV, FUS), including both FUS monotherapy and SRT+FUS arms (Table 7, Fig 6), using $n = 200$ matched-seed realizations per arm. Because every arm shares common random seeds, we analyze all contrasts as paired differences (200 matched pairs) rather than independent-sample comparisons, with 95% confidence intervals from $B = 1,000$ paired bootstrap resamples (Table 6). At day 365, every tested contrast — untreated vs. FUS-alone, AAV vs. AAV+FUS, SRT vs. SRT+FUS, and SRT+AAV vs. tri-modal — yields a paired 95% CI that excludes zero, meaning FUS’s effect is statistically detectable at this

sample size; the AAV vs. AAV+FUS and SRT+AAV vs. tri-modal contrasts are small and *negative* (paired mean differences of -0.032 and -0.028 nmol/g respectively, against baseline outcomes of order 20–25 nmol/g), while SRT vs. SRT+FUS is small and positive ($+0.013$ nmol/g). All four effects are two to three orders of magnitude smaller than the effect of AAV or SRT themselves. We formalize practical negligibility with a pre-defined threshold,

$$\delta_{\text{practical}} = 0.01 \times Y_{\text{untreated}}, \quad (1)$$

i.e. 1% of the untreated-arm outcome (day-365 untreated mean 250.76 nmol/g, so $\delta_{\text{practical}} = 2.51$ nmol/g). Every paired FUS contrast in Table 6 has $|\Delta| \ll \delta_{\text{practical}}$ (largest magnitude 0.032 nmol/g, 1.3% of the threshold itself). We therefore describe FUS’s day-365 contribution as statistically detectable but practically negligible, rather than as statistically indistinguishable from no effect: with $n = 200$ matched pairs the test has power to detect even a very small true effect, and does so, but by Eq (1) the effect’s magnitude is not therapeutically meaningful at this dose and schedule.

Table 6: **Paired matched-seed contrasts, day 365 (200 pairs, $B = 1,000$ bootstrap resamples).**

Contrast	Mean paired Δ (nmol/g)	95% CI	Within $\delta_{\text{practical}}$?
Untreated vs. FUS alone	$< 10^{-5}$	$(< 10^{-5}, < 10^{-5})$	Yes
AAV vs. AAV+FUS	-0.032	$(-0.045, -0.019)$	Yes
SRT vs. SRT+FUS	$+0.013$	$(0.010, 0.016)$	Yes
SRT+AAV vs. tri-modal	-0.028	$(-0.041, -0.017)$	Yes

$\delta_{\text{practical}} = 2.51$ nmol/g (Eq (1)). All four contrasts fall within the practical-equivalence margin despite three of the four having a 95% CI that excludes zero; statistical detectability and practical negligibility are reported separately here to avoid conflating them (see main text).

A Shapley-value decomposition of the tri-modal therapeutic benefit at day 365 attributes 94.88% of the total effect to AAV and 5.12% to SRT; FUS’s independent contribution is -0.0055% of total benefit (95% CI -0.008 to -0.003%). We do not interpret the sign of this value biologically; its magnitude is far below any threshold that would be practically meaningful relative to AAV’s or SRT’s contributions. Repeating this decomposition at three earlier time horizons (day 45, 90, 180) shows FUS’s contribution is not uniformly negligible across the treatment course: it is small but clearly positive early (2.14% of total benefit at day 45, 95% CI 2.08–2.20%), then decays through 0.014% (day 90) and 0.005% (day 180) to the small negative value at day 365 (Fig 5); every time point’s 95% CI excludes zero. This temporal pattern is consistent with, but does not by itself prove, the waning-AAV-expression dynamic identified independently in the trajectory data (Fig 4): a candidate mechanism is that FUS’s permeability boost confers a real, transient early advantage while AAV expression is still rising toward its peak (around day 250–300; Fig 4), and that this advantage fades as expression itself wanes, consistent with FUS’s only mechanistic channel of action (a multiplier on an entry rate that is, per the saturation analysis above, already saturated once AAV/SRT are active) having nothing further to contribute once the underlying process it multiplies has itself declined. We treat this as a model-specific interaction requiring further mechanistic verification rather than an established finding.

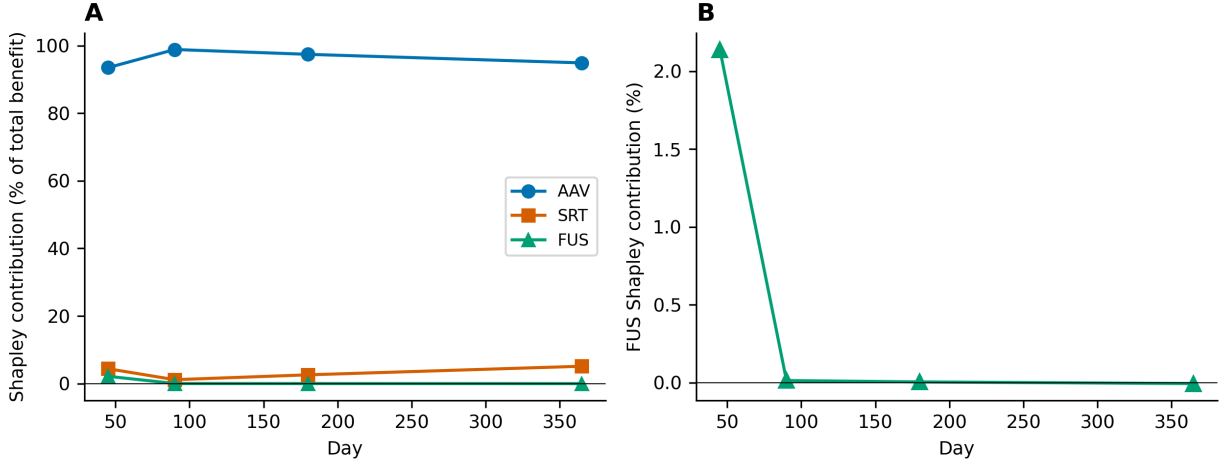


Figure 5: **FUS's Shapley contribution to tri-modal benefit declines and reverses sign over the treatment course.** (A) Shapley-attributed percentage of total tri-modal benefit for each intervention, at four time horizons (day 45, 90, 180, 365). (B) FUS's contribution alone with 95% bootstrap confidence intervals ($B = 1,000$): positive and statistically distinguishable from zero at day 45 (2.14%), declining through day 180, and small but significantly negative by day 365.

Table 7: **Full factorial design over the three constituent interventions (day-365 mean GM2 burden, nmol/g; $n = 200$ per arm).**

SRT	AAV	FUS	Mean burden $\pm$ SD
0	0	0	250.76 $\pm$ 135.00 (untreated)
0	0	1	250.76 $\pm$ 135.00 (FUS alone)
0	1	0	25.14 $\pm$ 18.41
0	1	1	25.17 $\pm$ 18.45
1	0	0	230.12 $\pm$ 126.10
1	0	1	230.11 $\pm$ 126.09
1	1	0	22.38 $\pm$ 16.33
1	1	1	22.41 $\pm$ 16.37 (tri-modal)

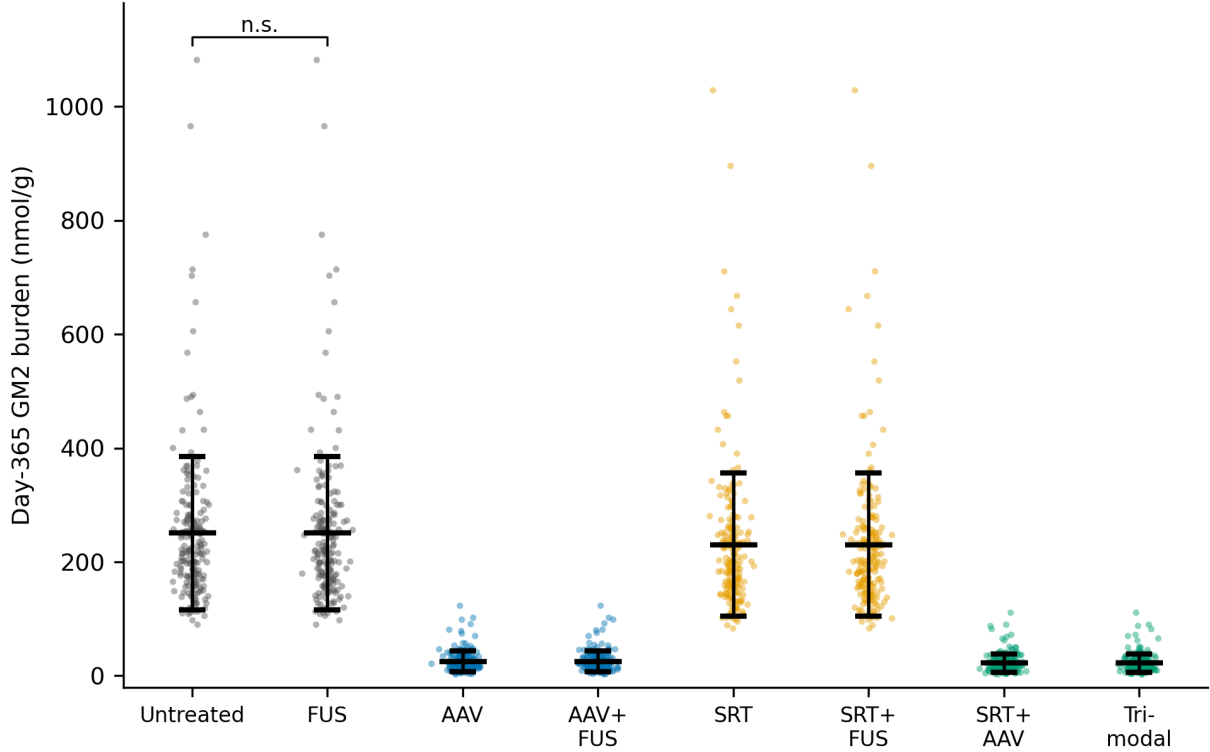


Figure 6: **Full 8-arm factorial design and paired FUS contrasts.** (A) Individual points show all $n = 200$ per-seed day-365 GM2 burden realizations for each of the eight intervention combinations (jittered horizontally for visibility; not summary statistics). Black markers with error bars denote the mean $\pm$ one standard deviation, computed from the same raw data shown. (B) The four paired matched-seed FUS contrasts from Table 6, shown as a forest plot (point estimate, 95% bootstrap CI) against the practical-equivalence band $|\Delta| < \delta_{\text{practical}}$ (Eq (1), shaded). All four contrasts fall well within the practical-equivalence band despite three of the four being statistically distinguishable from zero.

This pattern is mechanistically consistent with the saturation result above: FUS enters the model exclusively as a multiplicative gain on two rate constants governing AAV brain-entry and SRT brain-uptake. When AAV and SRT are both inactive, these rates are zero and the FUS multiplier has no effect regardless of its magnitude; when AAV or SRT are active, the corresponding rate constants are already saturated (Fig 2), so the FUS multiplier has essentially no remaining effect to confer.

Robustness of the FUS conclusion to a literature-bounded acoustic mechanism

Takeaway: FUS’s practical negligibility at day 365 is unchanged whether the model uses the calibrated adult FUS gain or a literature-derived, infant-permissive alternative.

The scalar FUS gain used throughout this paper (`fus_entry_gain_scale` = 1.5) was calibrated without reference to infant-specific acoustic biology. As a robustness check, we mapped the literature-bounded infant-permissive acoustic threshold identified in Limitations ($MI_0^{\text{infant}} \in [0.23, 0.46]$) onto this gain via the underlying MI-to-opening-probability relationship: at the model’s reference acoustic operating point (0.5 MHz, 0.6 MPa source pressure, 1% duty cycle, age 6 months), the adult threshold ($k = 1$) and the most permissive infant bound ($k = 2$) imply brain-region-averaged BBB-opening probabilities of 0.882 and 0.979 respectively — a ratio of 1.111, reflecting that the calibrated protocol already operates deep in the saturated part of the opening-probability curve even at the adult threshold. Scaling the calibrated gain’s departure from its no-FUS baseline by this ratio gives `fus_entry_gain_scale` = 1.666 for the permissive configuration (full derivation, including the rejected alternative scaling rule, archived at Data Availability). We reran

the complete 8-arm factorial design and the four-timepoint Shapley decomposition under this permissive configuration, with the adult configuration ($k = 1$) serving as an internal reproduction check of the original pipeline (its paired contrasts and Shapley values match those reported in Table 6 and the main text above to within Monte Carlo noise, confirming the two pipelines are computing the same quantity).

Under the permissive configuration, FUS’s day-45 contribution to the AAV contrast grows from 5.070 to 5.630 nmol/g, a 1.111-fold increase matching the derived opening-probability ratio to within 0.01%, confirming the gain mapping propagates through the full stochastic pipeline as intended rather than being lost to noise. At day 365, however, every paired contrast remains far below its practical-equivalence threshold under both configurations (largest magnitude: -0.035 nmol/g for AAV vs. AAV+FUS at $k = 2$, against $\delta_{\text{practical}} = 2.51$ nmol/g), and the day-365 FUS Shapley contribution remains a small negative value under both configurations (-0.0126 at $k = 1$; -0.0143 at $k = 2$, both 95% CIs excluding zero but three orders of magnitude below AAV’s 216.7 nmol/g contribution). Neither configuration changes this paper’s conclusion that FUS is statistically detectable but practically negligible by one year, and the two configurations agree with each other throughout rather than diverging.

We emphasize what this robustness check does and does not establish. It does not validate the acoustic model itself, which remains an exploratory, literature-bounded estimate resting on unproven species and mechanism analogies (Limitations). What it does establish is that this paper’s specific FUS conclusion is not an artifact of the particular calibrated gain value used: across the full literature-supported range for how much more permissive an infant BBB-opening threshold might plausibly be, the practical-negligibility finding holds.

Model verification and external validation

We verified five properties of the model’s implementation directly: that zero SRT dose leaves synthesis unmodified, that increasing SRT dose monotonically decreases synthesis (tested at four dose levels, ten seeds each), that the Milstein stochastic integration step reproduces analytic geometric Brownian motion moments for an isolated multiplicative-noise state, that FUS’s effect is negligible when AAV and SRT are both inactive (residual effect $< 10^{-3}$ relative to FUS’s effect when AAV or SRT are active), and that all 17 state variables remain within their defined bounds across 1,000 realizations, with lower-bound clamping disclosed explicitly where it occurs (most substantially for **inflammation**, which the unclamped integrator would drive negative in a majority of realizations). All five tests passed.

We additionally tested whether the as-documented ranking (Table 4) reflects assigned input width generally, rather than the saturation-specific mechanism proposed for $k_{T4,\text{entry}}$ specifically: across all nine parameters with nonzero sensitivity, the Spearman correlation between assigned relative width and total-order S_T under as-documented priors is weak and not statistically significant ($\rho = -0.31$, $p = 0.42$). This is consistent with our mechanistic account: $t_{\text{dose,total}}$ shares $k_{T4,\text{entry}}$ ’s wide assigned width (0.60 vs. 0.25 for the remaining seven parameters) but does not share its saturated functional response, and its S_T (0.079) sits between the enzyme-kinetics parameters and $k_{T4,\text{entry}}$ rather than tracking width directly. Width alone does not predict sensitivity in this model; the specific combination of wide assigned width and a saturated response curve does, for $k_{T4,\text{entry}}$ specifically.

Finally, we compared model output against three external sources: the untreated natural-history trajectory against published infantile GM2 natural history [?], the AAV-monotherapy response against a recent phase 1/2 dual-vector AAV gene-therapy trial [?], and the SRT-monotherapy response against the preclinical substrate-reduction-therapy literature. All three reveal substantive disagreements that we report directly rather than omit. The model’s untreated arm shows an early decline in the modeled functional score consistent with the literature’s early-regression pattern, but then spontaneously recovers above baseline by day 200–400 in every one of ten independent seeds, contradicting the literature’s monotonic, irreversible disease course; we traced this to a fixed linear decay term in the substrate-burden equation that dominates dynamics independent of treatment or residual enzyme activity, pulling substrate toward a lower quasi-steady-state even at zero enzyme activity. The model’s AAV-monotherapy arm shows near-complete, low-variance substrate clearance (99.5–99.6% reduction across ten seeds), substantially exceeding the 9–53% CSF substrate reduction reported in the cited clinical trial (whole-brain-tissue burden and CSF lipid-species concentration are not the same quantity, so we report this as a directional and order-of-magnitude comparison, not a formal error metric). The model’s SRT-monotherapy arm shows a real, seed-robust 51–63% mean brain substrate reduction, whereas

the closest available preclinical brain-tissue data for this drug class report the opposite direction in brain tissue specifically (substrate *increasing* to 120–150% of untreated, with reduction confined to peripheral organs and the drug’s benefit attributed to a non-substrate-reduction mechanism). We report all three disagreements as genuine, identified limitations of the underlying model’s absolute biological realism, distinct from the relative-sensitivity findings that are the focus of this paper: the Sobol, saturation, and factorial analyses above characterize which parameters and interventions drive variance and outcome differences *within this model as implemented*, and that characterization does not depend on the model’s absolute quantitative agreement with external data, but the disagreements documented here should inform how much weight is placed on this model’s absolute predictions independent of its relative-sensitivity structure.

Table 8 summarizes what is and is not established by the analyses above.

Table 8: **Model credibility summary.**

Credibility component	Current manuscript status	Interpretation
Numerical convergence	Sobol ranking stable by $N = 256$	Supports the sensitivity ranking
Prior-width robustness	Enzyme/synthesis parameters dominate under all three corrected prior specifications	Supports the reversal from the original claim
Saturation sweep	Entry-rate parameter insensitive over twelve orders of magnitude	Explains why the reversal is structurally expected
Factorial necessity	FUS’s terminal effect is statistically detectable but practically negligible	Separates parameter sensitivity from intervention effect
Code verification	Five implementation tests passed	Supports correctness of the implementation tested
External validation	Disagreements found in all three monotherapy arms tested	Limits confidence in absolute (not relative) predictions
Structural uncertainty	Not tested	Future work

To state plainly what this paper does and does not claim: this paper does not claim that BBB delivery is biologically unimportant, that FUS is ineffective *in vivo*, or that enzyme engineering should be clinically prioritized without experimental validation. It shows that, in this simulator and parameter regime, the original delivery-dominant Sobol conclusion is not robust to convergence, input-width, saturation, and factorial-necessity checks.

Discussion

Applied to this case, the four-part audit protocol overturns a sensitivity-derived conclusion using only convergence, prior-specification, and structural checks — no new experimental data. The central quantitative claim of the initial, as-documented analysis — that BBB-entry kinetics dominate GM2 outcome variance and that GM2 gangliosidosis is therefore delivery-limited rather than enzyme-limited — does not survive a prior-specification robustness check. Three independent lines of evidence converge on the same conclusion: a structural saturation mechanism explaining why the original result could not have been robust in principle (Fig 2); consistent dominance of enzyme-kinetics parameters across three independent prior specifications (Table 4); and generalization of this pattern across eight additional lysosomal storage disorders modeled with the same architecture (Table 5).

We emphasize that this reversal does not imply BBB delivery is biologically inconsequential. The saturation sweep (Fig 2) demonstrates that delivery is necessary — its complete removal reverts the model to the untreated state — but that, at the currently calibrated AAV dose, delivery is already sufficient to the point of saturation, such that further refinement of delivery kinetics specifically cannot improve outcomes in this model. Sensitivity and necessity are therefore not in conflict; a mechanism already operating at ceiling has no marginal sensitivity to its own further optimization without thereby being dispensable.

This interpretation — that further therapeutic gains in this model are enzyme-kinetics-limited rather than delivery-limited — carries translational implications, tempered by the external-validation findings above. Within this simulator, the corrected sensitivity ranking prioritizes enzyme/synthesis parameters over

BBB-entry kinetics. Because external validation identifies absolute-prediction failures in all three monotherapy arms tested, this should be interpreted as a model-audit result and a hypothesis for experimental follow-up — for example, directed evolution of HEXA/HEXB for improved catalytic capacity or reduced Michaelis constant may merit prioritized experimental attention relative to further delivery-system optimization — and not as a validated therapeutic prioritization claim.

More broadly, this analysis illustrates a general risk in stochastic biological world models calibrated from heterogeneous, evidence-tiered literature sources: a parameter’s assigned uncertainty, rather than its biological importance, can determine its apparent dominance in a variance-based sensitivity analysis. We suggest that the full four-part protocol demonstrated here — convergence verification, the three-way robustness check (as-documented, equal relative width, and promotion of the least-certain parameter to the calibration quality of the best-supported parameters), structural saturation analysis, and factorial necessity testing — offers a general, low-cost diagnostic that can be applied to any similarly calibrated model prior to drawing mechanistic conclusions from a single Sobol analysis.

Limitations

Several limitations should be considered. First, the archived simulator (Data Availability) includes a two-dimensional reaction–diffusion spatial module and FUS delivery controller, but all analyses in this paper were conducted on its lumped, non-spatial configuration; we have not verified whether the spatial module represents FUS’s regionally targeted mechanism of action (transient, localized BBB opening) or applies an equivalent permeability enhancement uniformly across the modeled tissue volume. All FUS-related conclusions in this paper therefore apply only to the scalar-gain, lumped implementation actually analyzed here, not to a spatially resolved acoustic delivery model. Second, the present analysis addresses parameter- and intervention-level robustness but does not address structural model uncertainty (i.e., whether the underlying functional forms themselves, such as Michaelis–Menten kinetics for enzyme turnover, are appropriate); this remains outside the scope of the present paper and should be evaluated in future work using alternative functional forms and nonparametric consistency checks. Third, the FUS mechanism is represented in the present model as a scalar multiplicative gain on entry rates; this paper therefore does not model acoustic field propagation, microbubble cavitation, regional pressure deposition, or FUS-specific safety constraints, and the finding that FUS shows negligible marginal benefit in this model should not be read as a claim about focused-ultrasound delivery in general. Fourth, and most substantively: the external validation reported above found real, seed-robust disagreements between this model’s absolute predictions and published data for all three monotherapy arms tested (untreated natural history, AAV monotherapy, and SRT monotherapy). These disagreements do not undermine the relative-sensitivity findings that are this paper’s central contribution — which parameters and interventions drive variance and outcome differences within this model as implemented is a well-posed question independent of the model’s absolute biological accuracy — but they are a material limitation on this model’s use for absolute quantitative prediction, and readers should weight the sensitivity-ranking and necessity findings above accordingly rather than treating this model’s absolute outputs as validated. Fifth, the FUS implementation does not include peak-negative pressure, mechanical index, acoustic field propagation, or microbubble dose; it uses a scalar multiplicative gain envelope applied to entry/uptake rates, with the schedule stated in Methods (three one-hour sessions on days 30, 60, and 90). Published microbubble-facilitated FUS-mediated AAV delivery work demonstrates the biophysical mechanism (transient, spatially localized BBB permeabilization) that this scalar abstraction omits entirely [?]. The findings in this paper therefore apply only to this scalar-gain abstraction and not to a spatially resolved acoustic-FUS protocol.

We separately explored, outside the analysis reported in this paper, what a literature-calibrated acoustic BBB-opening model would require. That exploration found no direct measurement of FUS-induced BBB-opening dose-response in infant, neonatal, or open-fontanelle tissue at any species, despite a broadened search across indexed literature, preprints, and registered clinical trials. It did find real, general (non-FUS-specific) evidence that BBB permeability maturation is size-dependent: neonatal mouse BBB permeability to small-molecule tracers is elevated relative to adult (a 2.0-fold ratio for one probe), but permeability to a macromolecular tracer (bovine serum albumin, size-comparable to AAV) is not detectably different [?], alongside a corrected primary-sourced adult mechanical-index threshold of $MI_0 = 0.46$ (95% CI 0.42–0.50) for BBB-opening [?]. Applying the permeability ratio as an explicit, stated (not validated) proportional scaling of this threshold gives a bounded range $MI_0^{\text{infant}} \in [0.23, 0.46]$ — as permissive as twice the adult

sensitivity, down to no different from adult — rather than a single value, since the underlying evidence is genuinely bidirectional by tracer size and collapsing it to one number would hide that disagreement. This bound rests on a species leap (mouse to human) and a mechanism leap (passive tracer diffusion to forced mechanical cavitation), both unproven; it is not a validated calibration of the acoustic biology itself. We did use this bound, mapped onto the calibrated FUS gain, to recompute the factorial and Shapley analyses in a separate model configuration and report the result as a robustness check (Results, “Robustness of the FUS conclusion...”); this does not validate the acoustic model, only tests whether this paper’s FUS conclusion depends on the specific calibrated gain value. The exact software package and version used for the Sobol/Shapley computation, and the specific natural-history sources used to set each disease’s evaluation window $t_{\max}$, are documented in the archived code and configuration files cited in Data Availability rather than in the main text.

Methods

Overview of the audit protocol

The four-part protocol introduced in the Introduction is implemented as follows: convergence verification (Sobol sample-size selection, defined below and reported in Results) precedes all other steps; prior-specification robustness compares three independent input-uncertainty scenarios, defined in “Prior-specification robustness protocol” below; structural saturation analysis is a continuous parameter sweep with no uncertainty-scenario dependence, defined in Results directly; and factorial necessity testing is the 2^3 intervention design defined in “Full factorial design and Shapley decomposition” below. Each is a generic step applicable to any similarly calibrated stochastic model; the GM2-specific configuration in which each is applied here is given in the subsections that follow.

Tri-modal treatment protocol

The tri-modal therapy evaluated throughout this paper (Introduction) consists of three concurrently administered interventions: substrate-reduction therapy (SRT), a single intravenous administration of AAV vector at the model’s calibrated full dose ($t_{\text{dose, total}} = 1.0$ in the model’s normalized dose units; see Results, Section “A structural saturation effect...”), and focused-ultrasound (FUS)-mediated transient BBB opening, administered as three sessions on days 30, 60, and 90 (the first coinciding with AAV administration), each lasting one hour with a 0.5-hour Gaussian edge-smoothing constant. FUS is represented in `gm2_model_v2.py` entirely as a dimensionless multiplicative gain envelope applied to two rate constants (BBB entry via `fus_entry_gain_scale`, calibrated value 1.5, and SRT brain uptake via a separate `fus_uptake_gain_scale` parameter, calibrated value 1.0, held fixed throughout and not included in the Sobol analysis); there is no peak-negative-pressure, mechanical-index, or microbubble-dose parameter anywhere in this pipeline, consistent with the scalar-gain limitation already noted (Limitations). All eight factorial arms (Table 7) share this same underlying protocol configuration, activating or deactivating each of the three interventions independently per arm.

Base model and computational environment

All analyses were conducted using our 17-state stochastic world model of GM2 gangliosidosis therapy (`gm2_model_v2.py`, a fixed forward simulator whose implementation is archived at the Zenodo record cited in Data Availability; SHA-256 file hash `6693427fa909a0a9...`, truncated to 16 hex characters — no version-control system was used for this project at any point, so a commit hash does not exist and this file hash is the only available integrity check), executed without modification to its core dynamics. Sobol and Shapley indices were computed with SALib [?] 1.5.2, Python 3.9.7, NumPy 2.0.2, and SciPy 1.13.1. These versions are inferred with high but not certain confidence ($\sim 90\%$, by the authors’ own assessment) from the single Python environment on the compute filesystem capable of running the analysis scripts, cross-referenced against package installation timestamps and per-job execution dates; no environment manifest (`requirements.txt`, `environment.yml`, or equivalent) was saved at the time the original jobs ran, so this could not be independently verified against a record contemporaneous with those runs. Intervention-level Shapley values (SRT/AAV/FUS, reported throughout Results) were computed with a custom, in-repository exact closed-form implementation

over the three binary interventions; a separate, permutation-based parameter-level Shapley-effects estimator following Song, Nelson, and Staum (2016) [?] was used independently for a Niemann–Pick C cross-check not otherwise reported in this manuscript. All Sobol, Shapley, and factorial analyses were executed on the Lonestar6 system at the Texas Advanced Computing Center (TACC), University of Texas at Austin, using SLURM array jobs to parallelize independent disease- and scenario-level evaluations across nodes. One early, superseded $N = 64$ convergence run references Vista’s GPU partitions in its submission script, but no Vista-specific software environment remains reconstructable on this filesystem; every result actually reported in this manuscript traces to a Lonestar6 job.

Global sensitivity analysis: definitions and estimator

We use variance-based (Sobol) global sensitivity indices, which decompose the total output variance $\text{Var}(Y)$ of a scalar model output Y (here, terminal GM2 burden at day 365) into contributions attributable to each input parameter θ_i and their interactions, following the Hoeffding–Sobol functional ANOVA decomposition [?]. For a generic input θ_i with complementary input set $\theta_{\sim i}$ (all other parameters), the first-order sensitivity index is defined as

$$S_i = \frac{\text{Var}_{\theta_i}(\mathbb{E}_{\theta_{\sim i}}[Y \mid \theta_i])}{\text{Var}(Y)}, \quad (2)$$

quantifying the fraction of output variance attributable to θ_i alone (additive effects). The total-order index,

$$S_{T,i} = 1 - \frac{\text{Var}_{\theta_{\sim i}}(\mathbb{E}_{\theta_i}[Y \mid \theta_{\sim i}])}{\text{Var}(Y)}, \quad (3)$$

captures all contributions from θ_i , including higher-order interactions with other parameters [?], and is the quantity reported throughout this paper (denoted S_T or S_T). We estimate Eq (3) using the Saltelli extension of the Sobol estimator [?], which requires two independent sampling matrices A and B (dimensions $N \times D$, where D is the number of parameters in the sampled problem definition and N is the base sample size) and D additional matrices $A_B^{(i)}$, each formed by replacing the i -th column of A with the corresponding column of B . Estimating total-order indices alone for all D parameters (SALib’s `calc_second_order=False` scheme) therefore requires

$$k = N(D + 2) \quad (4)$$

total model evaluations [?]; estimating second-order interaction indices simultaneously (with `calc_second_order=True`) instead requires an additional D matrices, giving

$$k = N(2D + 2) \quad (5)$$

total model evaluations. The full model problem definition includes $D = 14$ parameters (Table 3); the cross-disease analysis (Table 5) was run using Eq (4) (base sample sizes ranging $N = 128$ – $1,024$ across the ten disease parameterizations, i.e. $k = 16N$ evaluations per disease). The GM2 three-way prior-width comparison (Table 4) and its associated convergence check (Fig 2) were likewise run over the full 14-parameter set, of which nine parameters have nonzero sensitivity (Table 3); the convergence grid (Fig 2A) used $N = 64$ – $1,024$ with Eq (4) ($k = 16N$), while the production GM2 three-way analysis used $N = 256$ with Eq (5) ($k = N(2D + 2) = 30N = 7,680$ evaluations), sourced directly from the SLURM submission script and Sobol driver invocation for that run.

Convergence and uncertainty quantification

Sobol indices are known to converge slowly and are sensitive to under-sampling, particularly for total-order estimates of low-to-moderate-influence parameters [?]. We therefore treat convergence as a prerequisite to interpretation rather than an incidental diagnostic. We define convergence, following comparable practice in the systems-biology sensitivity-analysis literature [?], as the base sample size N beyond which (i) the rank ordering of parameters by S_T no longer changes and (ii) the bootstrap-estimated relative error of each reported index falls below 5%. Results of this convergence assessment for the GM2 model are reported in Results (Section “Convergence is a prerequisite for interpreting BBB-entry sensitivity”).

Prior-specification robustness protocol

We use the term *prior* throughout this paper in the global sensitivity analysis sense — an input uncertainty distribution assigned to an uncertain model parameter prior to sampling — and not in the Bayesian sense of a posterior distribution inferred from a likelihood; no Bayesian inference is performed anywhere in this analysis.

Three prior specifications were compared for each total-order Sobol analysis, holding the estimator, sample size, and random seed fixed across scenarios to isolate the effect of prior width alone: (i) *as-documented*, using each parameter’s reconstructed evidence-tiered prior width (Introduction); (ii) *equal relative width*, in which every parameter’s prior was rescaled to a common coefficient of variation $c = 0.144$ ($\pm 25\%$ uniform half-width, i.e. the same width already used for Tier-1 parameters in the as-documented scenario, applied uniformly to all 14 parameters regardless of tier); and (iii) *Tier-1-promoted*, in which only the BBB-entry parameter’s prior width was narrowed to match the width used for Tier-1 (multi-source) parameters, with all other priors held at their as-documented widths.

Cross-disease extension

All three prior specifications were applied to eight additional lysosomal storage disorders using the same model architecture with disease-specific synthesis, threshold, and enzyme-kinetic parameter substitutions. Evaluation windows were matched to each disease’s characteristic natural-history timescale ($t_{\max}$ ranging 150 d for Pompe to 8,213 d for Fabry; Table 5); the per-disease natural-history sources used to set $t_{\max}$ are documented in the archived configuration files rather than restated per-disease here (see Limitations). Base sample sizes N were adjusted per disease subject to computational budget constraints (Table 5); per Eq (4) with $D = 14$, this corresponds to $k = 16N$ total model evaluations per disease (e.g., $N = 1,024$ for Pompe, Krabbe, and GM1 corresponds to $k = 16,384$ evaluations each, confirmed directly from the submitted raw output files). At this full sample size and parameter set, Niemann–Pick C’s dominant total-order estimate is $S_T = 0.959$ [0.84, 1.00*], correctly bounded below the theoretical maximum of 1.0 as displayed (see Table 5 for the truncation convention).

Full factorial design and Shapley decomposition

All eight combinations of the three binary intervention indicators (SRT, AAV, FUS on/off) were evaluated under the model’s standard tri-modal dosing protocol, using $n = 200$ realizations per arm under matched random seeds (Table 7). The decomposition was repeated at four time horizons (day 45, 90, 180, and 365) to characterize the time-course of each intervention’s contribution (Fig 5). An intervention-level Shapley-value decomposition [?, ?] was computed over the three interventions: for each of the $3! = 6$ orderings of {SRT, AAV, FUS}, the marginal contribution of each intervention was computed as the change in terminal GM2 burden upon its addition to the subset of interventions already active in that ordering, and the Shapley value for each intervention is the average of its marginal contribution across all six orderings. This decomposition attributes the full tri-modal therapeutic effect exactly (Shapley values sum to the total effect by construction) and is more defensible than a pairwise Bliss-independence index, which assumes multiplicative independence between interventions and does not by itself test whether an intervention contributes a statistically distinguishable effect at all.

Data availability statement

Model-verification, external-validation, and width-sensitivity analysis outputs (Table 2 and the “Model verification and external validation” subsection) are archived at Zenodo, DOI: 10.5281/zenodo.21985843 (GitHub release v1.0.1 of https://github.com/SynapticOverflow/gm2_outputs, CC-BY 4.0). Whether this archive also contains `gm2_model_v2.py`, the Sobol/Shapley/factorial driver scripts, and the raw per-disease Sobol, trajectory, and factorial outputs underlying Tables 4–7 has not been independently confirmed for this manuscript; these materials will be added to the archive and this statement updated accordingly before submission if not already present. No new human, animal, or identifiable participant data were generated or used.

Ethics statement

This study used no new human participants, animals, or identifiable private data. All results are produced by simulation from literature-calibrated parameters and computational analysis of a previously implemented model. The work should not be interpreted as a clinical recommendation or therapeutic protocol.

Author contributions

Conceptualization: C.B. Methodology: K.N. Software: K.N. Formal analysis: K.N. Investigation: K.N. Writing – original draft: K.N. Writing – review & editing: K.N., C.B. Supervision: C.B.

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References

Supporting information

S1 Fig. Distribution of raw model outputs across the Saltelli sample matrix, all ten disease parameterizations. Histograms with kernel density estimates of the terminal GM2-analog burden (Y) sampled at each of the $k = N(D + 2)$ evaluation points underlying the cross-disease Sobol analysis (Table 5), confirmed directly from the submitted raw output files for each disease.

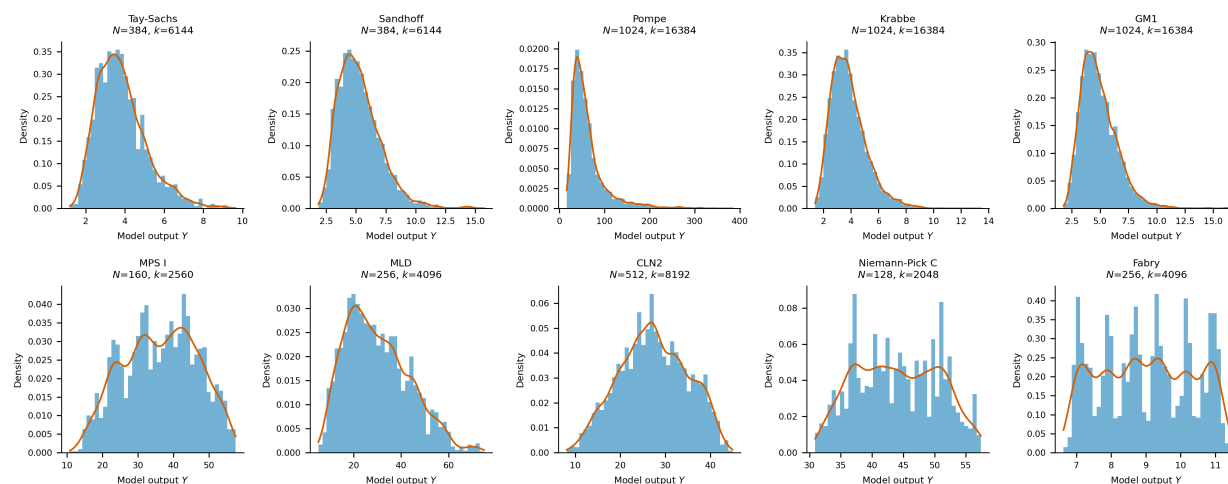


Figure 7: See S1 Fig description above.

S2 Fig. Full treated-arm disease-specific substrate burden trajectories with zoomed stochastic-noise insets, all ten disease parameterizations. Real time-course output (confirmed directly from the submitted trajectory files; $\Delta t = 0.1$ d, consistent with the model’s stated Milstein integration step) for brain GM2 burden under active treatment, extending to approximately twice each disease’s Sobol evaluation window $t_{\max}$. Each panel includes a zoomed inset revealing genuine per-timestep stochastic fluctuation (point-to-point noise confirmed directly from the raw data, not a rendering artifact of the compressed main axis). This figure appears as Fig 4 in the main Results.

S1 Table. Parameter provenance: central values, prior widths, and evidence tiers for all 14 model parameters. All priors are uniform distributions (SALib default; no log-uniform or other distribution is used anywhere in this pipeline). Tier 1 parameters use a $\pm 25\%$ half-width (CV 0.144); Tier 2 parameters use a $\pm 60\%$ half-width (CV 0.346). Both half-widths are a reconstruction of the original evidence-tiering scheme (Introduction), not values independently re-derived from the cited literature sources for this manuscript. Source counts (number of independent datasets informing each parameter) are not recorded as an explicit field anywhere in the archived configuration and are omitted rather than estimated.

Parameter	Central value	Half-width	CV	Evidence tier
Synthesis rate (g_{synth})	2.3	25%	0.144	1
Catalytic decay ($k_{T4,\text{decay}}$)	0.00417	25%	0.144	1
Catalytic capacity ($V_{\text{max},B}$)	12.5	25%	0.144	1
Michaelis constant ($K_{m,B}$)	75	25%	0.144	1
Expression rate ($k_{T4,\text{payload}}$)	0.08	25%	0.144	1
Total AAV dose ($t_{4\text{dose},\text{total}}$)	1.0	60%	0.346	2
BBB-entry rate ($k_{T4,\text{entry}}$)	0.06	60%	0.346	2
IC50	50	25%	0.144	1
FUS permeability gain (<code>fus_entry_gain_scale</code>)	1.5	25%	0.144	1
Infection threshold	650	60%	0.346	2
Infection kinetics	0.2	25%	0.144	1
Resistance kinetics	0.05	25%	0.144	1
Glial density	0.001	60%	0.346	2
Immune infiltration	0.01	60%	0.346	2